Md Shariful Islam Rayhan

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GitHub: github.com/srayhan706
Portfolio: rayhan706.vercel.app

Education

- **University of Dhaka** — Dhaka, Bangladesh
  Bachelor of Science; CGPA: 3.39/4 — *Jan 2022 - Jan 2026*
- **Notre Dame College** — Dhaka, Bangladesh
  HSC — *July 2018 - July 2020*

Skills Summary

- **Languages**: Python, C, C++,C#, Java, JavaScript, SQL
- **Frameworks**: React, Next.js, FastAPI, Flask, Android
- **Libraries & ML**: Pandas, NumPy, Scikit-learn, Data Mining
- **Tools**: Docker, Git, Bash, LaTeX
- **Platforms**: Linux, Web, Windows

Competitive Programming

LeetCode Profile: rayhansharif — **Rating (Max: 1814)**
Codeforces Profile: sharifrayhan — **Rating (Max: 1037)**

Projects

1. **CSEDU Department Website** *GitHub*
   Python, FastAPI, Javascript, NextJS
   A collaborative project to develop the official website of the University of Dhaka's CSE department, featuring user management, academic resources, announcements, and event scheduling.
   - Designed and implemented REST APIs for meetings, events, and announcements modules.
   - Developed the meetings, events, and announcements pages and integrated them with the backend APIs.

2. **Marketplace Web Application** *GitHub*
   Python, Flask, Javascript, NextJS
   Flask-based backend system with REST APIs designed using a modular and scalable architecture for efficient data handling.
   - Designed and implemented RESTful APIs for core marketplace functionalities.
   - Structured the backend using a modular architecture to ensure scalability and maintainability.
   - Handled data processing and routing efficiently for seamless client-server communication.

3. **Bookly — Android Application** *GitHub*
   Java, Android SDK
   Android application developed for online book purchasing with features such as search, user ratings, and contact modules.
   - Developed core UI screens for browsing, searching, and viewing book details.
   - Implemented user rating and feedback functionality to enhance user interaction.
   - Integrated contact and navigation modules for improved usability.

4. **NetConnect — Terminal-Based Group Chat System** *GitHub*
   Python, Socket Programming
   Multi-user real-time chat application with group creation, member approval, and access control using a custom networking protocol.
   - Implemented real-time multi-user messaging using socket programming.
   - Designed group creation and approval-based access control mechanisms.
   - Developed a custom communication protocol for efficient and reliable data exchange.

Thesis

CATS: Curvature-Adaptive Temporal Suppression for Long-Video Question Answering
A novel budget-aware frame selection method for multimodal large language models, addressing inefficiencies in long-video QA caused by suboptimal frame sampling. Introduces a curvature-adaptive temporal suppression mechanism to reduce redundancy while preserving both short-term salient events and long-range context.